Practice Activity 1

Problems and detailed worked solutions on Expected Value, Maximin, and Maximax.

Contents

- Problem 1 — Local Café Investment
- Problem 2 — Green Energy Production Mix
- Problem 3 — Global Shipping Network Design
- Problem 4 — Community Bakery Lunch Boxes
- Problem 5 — Rural Internet Subscriptions
- Problem 6 — Coffee Export Routes
- Problem 7 — Lake Ferry Tourism
- Problem 8 — Cold-Storage Contracts
- Problem 9 — Solar Installation Bids
- Problem 10 — Air Cargo Charters
- Problem 11 — Seafood Export Licenses

Problem 1 — Local Café Investment

- Problem description
- (C2) Interpreting decision alternatives, events, consequences, and states
- (C3) Building the payoff table
- (C4) Applying the Maximax criterion
- (C5) Applying the Maximin criterion
- (C1) Formulating the decision problem for an optimal choice using expected value

Problem description

A small café is deciding whether to launch a new line of artisan desserts. The transaction is daily café sales, where profit is earned from each dessert plate sold.

The owner has two options:

• **Launch the dessert line**, with profit outcomes depending on foot traffic: – High traffic → 42 thousand USD – Low traffic → 8 thousand USD

• **Keep the current menu**, with profit outcomes: – High traffic → 28 thousand USD – Low traffic → 20 thousand USD

Foot-traffic probabilities: • High traffic → 0.55 • Low traffic → 0.45

Question: Make a decision on the best option based on three perspectives: safe behavior (Maximin), balanced behavior (Expected Value), and risky behavior (Maximax).

Solution

Structured solution (C2) Interpreting decision alternatives, events, consequences, and states +

Element	Description
Alternatives	Launch dessert line; Keep current menu
States of nature	High traffic (0.55) Low traffic (0.45)
Events	Actual foot traffic after the decision
Consequences	Profit in thousand USD for each alternative–state pair

(C3) Building the payoff table +

Alternative	High traffic (0.55)	Low traffic (0.45)
Launch dessert line	42	8
Keep menu	28	20

(C4) Applying the Maximax criterion +

The maximax rule selects the alternative with the highest possible payoff.

$$\begin{aligned}\max_i \text{Payoff}(\text{Launch}, S_i) &= \max\{42, 8\} = 42\\ \max_i \text{Payoff}(\text{Keep}, S_i) &= \max\{28, 20\} = 28\end{aligned}$$

$$\max\{42, 28\} = 42$$

Because 42 is the largest best payoff, the maximax (risk-seeking) recommendation is *launch the dessert line*.

(C5) Applying the Maximin criterion +

The maximin rule focuses on the worst-case payoff of each alternative.

$$\begin{aligned}\min_i \text{Payoff}(\text{Launch}, S_i) &= \min\{42, 8\} = 8\\ \min_i \text{Payoff}(\text{Keep}, S_i) &= \min\{28, 20\} = 20\end{aligned}$$

$$\max\{8, 20\} = 20$$

Since 20 exceeds 8, the maximin (risk-averse) choice is *keep the current menu*.

(C1) Formulating the decision problem for an optimal choice using expected value +

We choose the option that maximizes the café's profit objective. Alternatives are **launch** or **keep** the menu; states are **high** or **low traffic**. Expected values:

$$\begin{aligned} \mathrm{EV}_{\mathrm{Launch}} &= 0.55(42) + 0.45(8) = 23.1 + 3.6 = 26.7 \\ \mathrm{EV}_{\mathrm{Keep}} &= 0.55(28) + 0.45(20) = 15.4 + 9 = 24.4 \end{aligned}$$

$$\max\{26.7, 24.4\} = 26.7$$

Summary of recommendations across decision criteria

- Maximax favors *launching the dessert line* (highest payoff of 42).
- Maximin favors *keeping the current menu* (best worst-case payoff of 20).
- Expected value favors *launching the dessert line* (EV of 26.7).

Launching desserts leads on optimistic and average outcomes, while the current menu protects the downside.

(C2) Interpreting decision alternatives, events, consequences, and states

Element	Description
Alternatives	Launch dessert line; Keep current menu
States of nature	High traffic (0.55) Low traffic (0.45)
Events	Actual foot traffic after the decision
Consequences	Profit in thousand USD for each alternative–state pair

(C3) Building the payoff table

Alternative	High traffic (0.55)	Low traffic (0.45)
Launch dessert line	42	8
Keep menu	28	20

(C4) Applying the Maximax criterion

The maximax rule selects the alternative with the highest possible payoff.

$$\begin{aligned} \max_i \mathrm{Payoff}(\mathrm{Launch}, S_i) &= \max\{42, 8\} = 42 \\ \max_i \mathrm{Payoff}(\mathrm{Keep}, S_i) &= \max\{28, 20\} = 28 \end{aligned}$$

$$\max\{42, 28\} = 42$$

Because 42 is the largest best payoff, the maximax (risk-seeking) recommendation is *launch the dessert line.*

(C5) Applying the Maximin criterion

The maximin rule focuses on the worst-case payoff of each alternative.

$$\begin{aligned}\min_i \text{Payoff}(\text{Launch}, S_i) &= \min\{42, 8\} = 8\\ \min_i \text{Payoff}(\text{Keep}, S_i) &= \min\{28, 20\} = 20\end{aligned}$$

$$\max\{8, 20\} = 20$$

Since 20 exceeds 8, the maximin (risk-averse) choice is *keep the current menu*.

(C1) Formulating the decision problem for an optimal choice using expected value

We choose the option that maximizes the café's profit objective. Alternatives are **launch** or **keep** the menu; states are **high** or **low traffic**. Expected values:

$$\begin{aligned}\text{EV}_{\text{Launch}} &= 0.55(42) + 0.45(8) = 23.1 + 3.6 = 26.7\\ \text{EV}_{\text{Keep}} &= 0.55(28) + 0.45(20) = 15.4 + 9 = 24.4\end{aligned}$$

$$\max\{26.7, 24.4\} = 26.7$$

Summary of recommendations across decision criteria

- Maximax favors *launching the dessert line* (highest payoff of 42).
- Maximin favors *keeping the current menu* (best worst-case payoff of 20).
- Expected value favors *launching the dessert line* (EV of 26.7).

Launching desserts leads on optimistic and average outcomes, while the current menu protects the downside.

Problem 2 — Green Energy Production Mix

- Problem description
- (C2) Interpreting decision alternatives, events, consequences, and states
- (C3) Building the payoff table
- (C4) Applying the Maximax criterion
- (C5) Applying the Maximin criterion
- (C1) Formulating the decision problem for an optimal choice using expected value

Problem description

A renewable-energy company must select an electricity generation mix. The transaction is annual electricity sales, where profit is earned per megawatt-hour generated and sold.

Three weather states are possible:

• Windy • Sunny • Cloudy

The company is comparing three strategies:

1. Build wind turbines 2. Build solar farms 3. Build a balanced hybrid system

Expected profits (in thousand USD):

• Wind turbines → 90 (Windy), 45 (Sunny), 30 (Cloudy) • Solar farms → 35 (Windy), 95 (Sunny), 25 (Cloudy) • Hybrid system → 70 (Windy), 65 (Sunny), 60 (Cloudy)

Weather probabilities: • Windy → 0.30 • Sunny → 0.45 • Cloudy → 0.25

Question: Make a decision on the best strategy based on three perspectives: safe behavior (Maximin), balanced behavior (Expected Value), and risky behavior (Maximax).

Solution

Structured solution (C2) Interpreting decision alternatives, events, consequences, and states +

Element	Description
Alternatives	Wind turbines; Solar farms; Hybrid system
States of nature	Windy (0.30) Sunny (0.45) Cloudy (0.25)
Events	Weather outcome affecting generation
Consequences	Profit in thousand USD for each strategy–weather combination

(C3) Building the payoff table +

Strategy	Windy (0.30)	Sunny (0.45)	Cloudy (0.25)
Wind turbines	90	45	30
Solar farms	35	95	25
Hybrid system	70	65	60

(C4) Applying the Maximax criterion +

Maximax compares the highest payoff for each alternative.

$$\begin{aligned}\max_i \text{Payoff}(\text{Wind}, S_i) &= \max\{90, 45, 30\} = 90\\ \max_i \text{Payoff}(\text{Solar}, S_i) &= \max\{35, 95, 25\} = 95\\ \max_i \text{Payoff}(\text{Hybrid}, S_i) &= \max\{70, 65, 60\} = 70\end{aligned}$$

$$\max\{90, 95, 70\} = 95$$

The largest best payoff is 95, so the maximax decision is *build solar farms*.

(C5) Applying the Maximin criterion +

Maximin compares the worst payoff for each alternative.

$$\begin{aligned}\min_i \text{Payoff}(\text{Wind}, S_i) &= \min\{90, 45, 30\} = 30\\ \min_i \text{Payoff}(\text{Solar}, S_i) &= \min\{35, 95, 25\} = 25\\ \min_i \text{Payoff}(\text{Hybrid}, S_i) &= \min\{70, 65, 60\} = 60\end{aligned}$$

$$\max\{30, 25, 60\} = 60$$

The maximin recommendation is the *hybrid system*.

(C1) Formulating the decision problem for an optimal choice using expected value +

The firm selects the generation mix that maximizes profit. Alternatives are **wind**, **solar**, or **hybrid** and the uncertain states are **windy**, **sunny**, or **cloudy**. Expected values:

$$\begin{aligned}\text{EV}_{\text{Wind}} &= 0.30(90) + 0.45(45) + 0.25(30) = 27 + 20.25 + 7.5 = 54.75\\ \text{EV}_{\text{Solar}} &= 0.30(35) + 0.45(95) + 0.25(25) = 10.5 + 42.75 + 6.25 = 59.5\\ \text{EV}_{\text{Hybrid}} &= 0.30(70) + 0.45(65) + 0.25(60) = 21 + 29.25 + 15 = 65.25\end{aligned}$$

$$\max\{54.75, 59.5, 65.25\} = 65.25$$

Summary of recommendations across decision criteria

- Maximax favors *solar farms* (highest payoff of 95).
- Maximin favors the *hybrid system* (best worst-case payoff of 60).
- Expected value favors the *hybrid system* (EV of 65.25).

Solar has the best upside, but the hybrid option balances downside protection with the strongest average profit.

(C2) Interpreting decision alternatives, events, consequences, and states

Element	Description
Alternatives	Wind turbines; Solar farms; Hybrid system
States of nature	Windy (0.30) Sunny (0.45) Cloudy (0.25)
Events	Weather outcome affecting generation
Consequences	Profit in thousand USD for each strategy–weather combination

(C3) Building the payoff table

Strategy	Windy (0.30)	Sunny (0.45)	Cloudy (0.25)
Wind turbines	90	45	30
Solar farms	35	95	25
Hybrid system	70	65	60

(C4) Applying the Maximax criterion

Maximax compares the highest payoff for each alternative.

$$\begin{aligned}\max_i \text{Payoff}(\text{Wind}, S_i) &= \max\{90, 45, 30\} = 90\\ \max_i \text{Payoff}(\text{Solar}, S_i) &= \max\{35, 95, 25\} = 95\\ \max_i \text{Payoff}(\text{Hybrid}, S_i) &= \max\{70, 65, 60\} = 70\end{aligned}$$

$$\max\{90, 95, 70\} = 95$$

The largest best payoff is 95, so the maximax decision is *build solar farms*.

(C5) Applying the Maximin criterion

Maximin compares the worst payoff for each alternative.

$$\begin{aligned}\min_i \text{Payoff}(\text{Wind}, S_i) &= \min\{90, 45, 30\} = 30\\ \min_i \text{Payoff}(\text{Solar}, S_i) &= \min\{35, 95, 25\} = 25\\ \min_i \text{Payoff}(\text{Hybrid}, S_i) &= \min\{70, 65, 60\} = 60\end{aligned}$$

$$\max\{30, 25, 60\} = 60$$

The maximin recommendation is the *hybrid system*.

(C1) Formulating the decision problem for an optimal choice using expected value

The firm selects the generation mix that maximizes profit. Alternatives are **wind**, **solar**, or **hybrid** and the uncertain states are **windy**, **sunny**, or **cloudy**. Expected values:

$$\begin{aligned}\text{EV}_{\text{Wind}} &= 0.30(90) + 0.45(45) + 0.25(30) = 27 + 20.25 + 7.5 = 54.75\\ \text{EV}_{\text{Solar}} &= 0.30(35) + 0.45(95) + 0.25(25) = 10.5 + 42.75 + 6.25 = 59.5\\ \text{EV}_{\text{Hybrid}} &= 0.30(70) + 0.45(65) + 0.25(60) = 21 + 29.25 + 15 = 65.25\end{aligned}$$

$$\max\{54.75, 59.5, 65.25\} = 65.25$$

Summary of recommendations across decision criteria

- Maximax favors *solar farms* (highest payoff of 95).
- Maximin favors the *hybrid system* (best worst-case payoff of 60).
- Expected value favors the *hybrid system* (EV of 65.25).

Solar has the best upside, but the hybrid option balances downside protection with the strongest average profit.

Problem 3 — Global Shipping Network Design

- Problem description
- (C2) Interpreting decision alternatives, events, consequences, and states
- (C3) Building the payoff table
- (C4) Applying the Maximax criterion
- (C5) Applying the Maximin criterion
- (C1) Formulating the decision problem for an optimal choice using expected value

Problem description

A multinational logistics firm is choosing between two shipping network designs for long-term freight contracts. The transaction is global freight services, where profit is earned per contract delivered on schedule.

Management is considering:

• **Centralized mega-hub network** • **Regional multi-hub network**

The firm models four trade conditions: • Trade boom (0.20) • Stable trade (0.35) • Moderate disruptions (0.30) • Severe disruptions (0.15)

Expected profits (in thousand USD) are: • Centralized network $\rightarrow 220, 140, 40, -30$ • Regional network $\rightarrow 180, 160, 110, 60$

Question: Make a decision on the best configuration based on three perspectives: safe behavior (Maximin), balanced behavior (Expected Value), and risky behavior (Maximax).

Solution

Structured solution (C2) Interpreting decision alternatives, events, consequences, and states +

Element	Description
Alternatives	Centralized mega-hub; Regional multi-hub
States of nature	Boom (0.20) Stable (0.35) Moderate disruptions (0.30) Severe disruptions (0.15)
Events	Trade conditions realized during the planning horizon
Consequences	Profit in thousand USD for each network–state pair

(C3) Building the payoff table +

Network	Boom (0.20)	Stable (0.35)	Moderate disruptions (0.30)	Severe disruptions (0.15)
Centralized mega-hub	220	140	40	-30
Regional multi-hub	180	160	110	60

(C4) Applying the Maximax criterion +

Maximax selects the alternative with the highest possible payoff.

$$\begin{aligned}\max_i \text{Payoff}(\text{Centralized}, S_i) &= \max\{220, 140, 40, -30\} = 220\\ \max_i \text{Payoff}(\text{Regional}, S_i) &= \max\{180, 160, 110, 60\} = 180\end{aligned}$$

$$\max\{220, 180\} = 220$$

The maximax choice is the *centralized mega-hub network*.

(C5) Applying the Maximin criterion +

Maximin compares the worst payoff for each alternative.

$$\begin{aligned}\min_i \text{Payoff}(\text{Centralized}, S_i) &= \min\{220, 140, 40, -30\} = -30\\ \min_i \text{Payoff}(\text{Regional}, S_i) &= \min\{180, 160, 110, 60\} = 60\end{aligned}$$

$$\max\{-30, 60\} = 60$$

The maximin recommendation is the *regional multi-hub network*.

(C1) Formulating the decision problem for an optimal choice using expected value +

The firm chooses the network that maximizes long-run profit. Alternatives are **centralized** or **regional** and the states are **boom**, **stable**, **moderate**, and **severe disruptions**. Expected values:

$$\begin{aligned}\text{EV}_{\text{Centralized}} &= 0.20(220) + 0.35(140) + 0.30(40) + 0.15(-30)\\ &= 44 + 49 + 12 - 4.5 = 100.5\\ \text{EV}_{\text{Regional}} &= 0.20(180) + 0.35(160) + 0.30(110) + 0.15(60)\\ &= 36 + 56 + 33 + 9 = 134\end{aligned}$$

$$\max\{100.5, 134\} = 134$$

Summary of recommendations across decision criteria

- Maximax favors the *centralized mega-hub network* (highest payoff of 220).
- Maximin favors the *regional multi-hub network* (best worst-case payoff of 60).
- Expected value favors the *regional multi-hub network* (EV of 134).

Centralization offers peak upside, but the regional option delivers stronger protection and higher average profit.

(C2) Interpreting decision alternatives, events, consequences, and states

Element	Description
Alternatives	Centralized mega-hub; Regional multi-hub
States of nature	Boom (0.20) Stable (0.35) Moderate disruptions (0.30) Severe disruptions (0.15)
Events	Trade conditions realized during the planning horizon
Consequences	Profit in thousand USD for each network–state pair

(C3) Building the payoff table

Network	Boom (0.20)	Stable (0.35)	Moderate disruptions (0.30)	Severe disruptions (0.15)
Centralized mega-hub	220	140	40	-30
Regional multi-hub	180	160	110	60

(C4) Applying the Maximax criterion

Maximax selects the alternative with the highest possible payoff.

$$\max_i \text{Payoff}(\text{Centralized}, S_i)\& = \max\{220, 140, 40, -30\} = 220$$

$$\max_i \text{Payoff}(\text{Regional}, S_i)\& = \max\{180, 160, 110, 60\} = 180$$

$$\max\{220, 180\} = 220$$

The maximax choice is the *centralized mega-hub network*.

(C5) Applying the Maximin criterion

Maximin compares the worst payoff for each alternative.

$$\min_i \text{Payoff}(\text{Centralized}, S_i)\& = \min\{220, 140, 40, -30\} = -30$$

$$\min_i \text{Payoff}(\text{Regional}, S_i)\& = \min\{180, 160, 110, 60\} = 60$$

$$\max\{-30, 60\} = 60$$

The maximin recommendation is the *regional multi-hub network*.

(C1) Formulating the decision problem for an optimal choice using expected value

The firm chooses the network that maximizes long-run profit. Alternatives are **centralized** or **regional** and the states are **boom**, **stable**, **moderate**, and **severe disruptions**. Expected

values:

$$\begin{aligned} \text{EV}_{\text{Centralized}} &= 0.20(220) + 0.35(140) + 0.30(40) + 0.15(-30) \\ &= 44 + 49 + 12 - 4.5 = 100.5 \\ \text{EV}_{\text{Regional}} &= 0.20(180) + 0.35(160) + 0.30(110) + 0.15(60) \\ &= 36 + 56 + 33 + 9 = 134 \end{aligned}$$

$$\max\{100.5, 134\} = 134$$

Summary of recommendations across decision criteria

- Maximax favors the *centralized mega-hub network* (highest payoff of 220).
- Maximin favors the *regional multi-hub network* (best worst-case payoff of 60).
- Expected value favors the *regional multi-hub network* (EV of 134).

Centralization offers peak upside, but the regional option delivers stronger protection and higher average profit.

Problem 4 — Community Bakery Lunch Boxes

- Problem description
- (C2) Interpreting decision alternatives, events, consequences, and states
- (C3) Building the payoff table
- (C4) Applying the Maximax criterion
- (C5) Applying the Maximin criterion
- (C1) Formulating the decision problem for an optimal choice using expected value

Problem description

A community bakery is bidding on office lunch-box contracts. The transaction is meal box sales, where profit is earned per lunch box delivered.

The bakery can choose one pricing strategy:

• **Standard packaging** with a profit of 0.04 thousand USD per lunch box • **Premium packaging** with a profit of 0.07 thousand USD per lunch box

Demand states: • High office demand (0.60) • Low office demand (0.40)

Expected number of lunch boxes: • Standard packaging → 900 (High), 550 (Low) • Premium packaging → 650 (High), 250 (Low)

Question: Compute profits and choose the best option using Maximin, Expected Value, and Maximax.

Solution

Structured solution (C2) Interpreting decision alternatives, events, consequences, and states +

Element	Description
Alternatives	Standard packaging; Premium packaging
States of nature	High demand (0.60) Low demand (0.40)
Events	Actual number of lunch-box contracts realized
Consequences	Profit in thousand USD, based on boxes × profit per box

(C3) Building the payoff table +

Profit calculations (thousand USD):

- Standard: High $\rightarrow 900 \times 0.04 = 36$; Low $\rightarrow 550 \times 0.04 = 22$.
- Premium: High $\rightarrow 650 \times 0.07 = 45.5$; Low $\rightarrow 250 \times 0.07 = 17.5$.

Alternative	High demand (0.60)	Low demand (0.40)
Standard packaging	36	22
Premium packaging	45.5	17.5

(C4) Applying the Maximax criterion +

Maximax compares the best outcomes.

$$\begin{aligned} \max_i \text{Payoff}(\text{Standard}, S_i) &= \max\{36, 22\} = 36 \\ \max_i \text{Payoff}(\text{Premium}, S_i) &= \max\{45.5, 17.5\} = 45.5 \end{aligned}$$

$$\max\{36, 45.5\} = 45.5$$

The maximax choice is *premium packaging*.

(C5) Applying the Maximin criterion +

Maximin compares the worst outcomes.

$$\begin{aligned} \min_i \text{Payoff}(\text{Standard}, S_i) &= \min\{36, 22\} = 22 \\ \min_i \text{Payoff}(\text{Premium}, S_i) &= \min\{45.5, 17.5\} = 17.5 \end{aligned}$$

$$\max\{22, 17.5\} = 22$$

The maximin recommendation is *standard packaging*.

(C1) Formulating the decision problem for an optimal choice using expected value +

We select the pricing strategy that maximizes expected profit. Expected values:

$$\begin{aligned} \text{EV}_{\text{Standard}} &= 0.60(36) + 0.40(22) = 21.6 + 8.8 = 30.4 \\ \text{EV}_{\text{Premium}} &= 0.60(45.5) + 0.40(17.5) = 27.3 + 7 = 34.3 \end{aligned}$$

$$\max\{30.4, 34.3\} = 34.3$$

Summary of recommendations across decision criteria

- Maximax favors *premium packaging* (highest payoff of 45.5).
- Maximin favors *standard packaging* (best worst-case payoff of 22).
- Expected value favors *premium packaging* (EV of 34.3).

Premium packaging leads on optimistic and average profits, while standard packaging limits downside risk.

(C2) Interpreting decision alternatives, events, consequences, and states

Element	Description
Alternatives	Standard packaging; Premium packaging
States of nature	High demand (0.60) Low demand (0.40)
Events	Actual number of lunch-box contracts realized
Consequences	Profit in thousand USD, based on boxes × profit per box

(C3) Building the payoff table

Profit calculations (thousand USD):

- Standard: High $\rightarrow 900 \times 0.04 = 36$; Low $\rightarrow 550 \times 0.04 = 22$.
- Premium: High $\rightarrow 650 \times 0.07 = 45.5$; Low $\rightarrow 250 \times 0.07 = 17.5$.

Alternative	High demand (0.60)	Low demand (0.40)
Standard packaging	36	22
Premium packaging	45.5	17.5

(C4) Applying the Maximax criterion

Maximax compares the best outcomes.

$$\begin{aligned} \max_i \text{Payoff}(\text{Standard}, S_i) &= \max\{36, 22\} = 36 \\ \max_i \text{Payoff}(\text{Premium}, S_i) &= \max\{45.5, 17.5\} = 45.5 \end{aligned}$$

$$\max\{36, 45.5\} = 45.5$$

The maximax choice is *premium packaging*.

(C5) Applying the Maximin criterion

Maximin compares the worst outcomes.

$$\begin{aligned}\min_i \text{Payoff}(\text{Standard}, S_i) &= \min\{36, 22\} = 22\\ \min_i \text{Payoff}(\text{Premium}, S_i) &= \min\{45.5, 17.5\} = 17.5\end{aligned}$$

$$\max\{22, 17.5\} = 22$$

The maximin recommendation is *standard packaging*.

(C1) Formulating the decision problem for an optimal choice using expected value

We select the pricing strategy that maximizes expected profit. Expected values:

$$\begin{aligned}\text{EV}_{\text{Standard}} &= 0.60(36) + 0.40(22) = 21.6 + 8.8 = 30.4\\ \text{EV}_{\text{Premium}} &= 0.60(45.5) + 0.40(17.5) = 27.3 + 7 = 34.3\end{aligned}$$

$$\max\{30.4, 34.3\} = 34.3$$

Summary of recommendations across decision criteria

- Maximax favors *premium packaging* (highest payoff of 45.5).
- Maximin favors *standard packaging* (best worst-case payoff of 22).
- Expected value favors *premium packaging* (EV of 34.3).

Premium packaging leads on optimistic and average profits, while standard packaging limits downside risk.

Problem 5 — Rural Internet Subscriptions

- Problem description
- (C2) Interpreting decision alternatives, events, consequences, and states
- (C3) Building the payoff table
- (C4) Applying the Maximax criterion
- (C5) Applying the Maximin criterion
- (C1) Formulating the decision problem for an optimal choice using expected value

Problem description

A rural internet provider is selecting a subscription plan mix. The transaction is monthly broadband subscriptions, where profit is earned per subscription sold.

Decision alternatives:

• **Basic plan** with profit of 0.03 thousand USD per subscription • **Plus plan** with profit of 0.05 thousand USD per subscription • **Pro plan** with profit of 0.08 thousand USD per subscription

Enrollment states: • High enrollment (0.55) • Low enrollment (0.45)

Expected subscriptions sold: • Basic → 1400 (High), 900 (Low) • Plus → 1000 (High), 700 (Low) • Pro → 600 (High), 300 (Low)

Question: Compute profits and choose the best plan using Maximin, Expected Value, and Maximax.

Solution

Structured solution (C2) Interpreting decision alternatives, events, consequences, and states +

Element	Description
Alternatives	Basic plan; Plus plan; Pro plan
States of nature	High enrollment (0.55) Low enrollment (0.45)
Events	Actual subscription uptake after plan selection
Consequences	Profit in thousand USD based on subscriptions × profit per subscription

(C3) Building the payoff table +

Profit calculations (thousand USD):

- Basic: High → $1400 \times 0.03 = 42$; Low → $900 \times 0.03 = 27$.
- Plus: High → $1000 \times 0.05 = 50$; Low → $700 \times 0.05 = 35$.
- Pro: High → $600 \times 0.08 = 48$; Low → $300 \times 0.08 = 24$.

Alternative	High enrollment (0.55)	Low enrollment (0.45)
Basic plan	42	27
Plus plan	50	35
Pro plan	48	24

(C4) Applying the Maximax criterion +

Maximax compares the best outcomes.

$$\begin{aligned}\max_i \text{Payoff}(\text{Basic}, S_i) &= \max\{42, 27\} = 42\\ \max_i \text{Payoff}(\text{Plus}, S_i) &= \max\{50, 35\} = 50\\ \max_i \text{Payoff}(\text{Pro}, S_i) &= \max\{48, 24\} = 48\end{aligned}$$

$$\max\{42, 50, 48\} = 50$$

The maximax choice is the *Plus plan*.

(C5) Applying the Maximin criterion +

Maximin compares the worst outcomes.

$$\begin{aligned}\min_i \text{Payoff}(\text{Basic}, S_i)\& &= \min\{42, 27\} = 27\\ \min_i \text{Payoff}(\text{Plus}, S_i)\& &= \min\{50, 35\} = 35\\ \min_i \text{Payoff}(\text{Pro}, S_i)\& &= \min\{48, 24\} = 24\end{aligned}$$

$$\max\{27, 35, 24\} = 35$$

The maximin recommendation is the *Plus plan*.

(C1) Formulating the decision problem for an optimal choice using expected value +

We select the plan mix that maximizes expected profit. Expected values:

$$\begin{aligned}\text{EV}_{\text{Basic}}\& &= 0.55(42) + 0.45(27) = 23.1 + 12.15 = 35.25\\ \text{EV}_{\text{Plus}}\& &= 0.55(50) + 0.45(35) = 27.5 + 15.75 = 43.25\\ \text{EV}_{\text{Pro}}\& &= 0.55(48) + 0.45(24) = 26.4 + 10.8 = 37.2\end{aligned}$$

$$\max\{35.25, 43.25, 37.2\} = 43.25$$

Summary of recommendations across decision criteria

- Maximax favors the *Plus plan* (highest payoff of 50).
- Maximin favors the *Plus plan* (best worst-case payoff of 35).
- Expected value favors the *Plus plan* (EV of 43.25).

The Plus plan balances a strong upside with the best downside protection and highest average profit.

(C2) Interpreting decision alternatives, events, consequences, and states

Element	Description
Alternatives	Basic plan; Plus plan; Pro plan
States of nature	High enrollment (0.55) Low enrollment (0.45)
Events	Actual subscription uptake after plan selection
Consequences	Profit in thousand USD based on subscriptions $\times$ profit per subscription

(C3) Building the payoff table

Profit calculations (thousand USD):

- Basic: High $\rightarrow 1400 \times 0.03 = 42$; Low $\rightarrow 900 \times 0.03 = 27$.
- Plus: High $\rightarrow 1000 \times 0.05 = 50$; Low $\rightarrow 700 \times 0.05 = 35$.
- Pro: High $\rightarrow 600 \times 0.08 = 48$; Low $\rightarrow 300 \times 0.08 = 24$.

Alternative	High enrollment (0.55)	Low enrollment (0.45)
Basic plan	42	27
Plus plan	50	35
Pro plan	48	24

(C4) Applying the Maximax criterion

Maximax compares the best outcomes.

$$\begin{aligned} \max_i \text{Payoff}(\text{Basic}, S_i) &= \max\{42, 27\} = 42 \\ \max_i \text{Payoff}(\text{Plus}, S_i) &= \max\{50, 35\} = 50 \\ \max_i \text{Payoff}(\text{Pro}, S_i) &= \max\{48, 24\} = 48 \end{aligned}$$

$$\max\{42, 50, 48\} = 50$$

The maximax choice is the *Plus plan*.

(C5) Applying the Maximin criterion

Maximin compares the worst outcomes.

$$\begin{aligned} \min_i \text{Payoff}(\text{Basic}, S_i) &= \min\{42, 27\} = 27 \\ \min_i \text{Payoff}(\text{Plus}, S_i) &= \min\{50, 35\} = 35 \\ \min_i \text{Payoff}(\text{Pro}, S_i) &= \min\{48, 24\} = 24 \end{aligned}$$

$$\max\{27, 35, 24\} = 35$$

The maximin recommendation is the *Plus plan*.

(C1) Formulating the decision problem for an optimal choice using expected value

We select the plan mix that maximizes expected profit. Expected values:

$$\begin{aligned} \text{EV}_{\text{Basic}} &= 0.55(42) + 0.45(27) = 23.1 + 12.15 = 35.25 \\ \text{EV}_{\text{Plus}} &= 0.55(50) + 0.45(35) = 27.5 + 15.75 = 43.25 \\ \text{EV}_{\text{Pro}} &= 0.55(48) + 0.45(24) = 26.4 + 10.8 = 37.2 \end{aligned}$$

$$\max\{35.25, 43.25, 37.2\} = 43.25$$

Summary of recommendations across decision criteria

- Maximax favors the *Plus plan* (highest payoff of 50).
- Maximin favors the *Plus plan* (best worst-case payoff of 35).
- Expected value favors the *Plus plan* (EV of 43.25).

The Plus plan balances a strong upside with the best downside protection and highest average profit.

Problem 6 — Coffee Export Routes

- Problem description
- (C2) Interpreting decision alternatives, events, consequences, and states
- (C3) Building the payoff table
- (C4) Applying the Maximax criterion
- (C5) Applying the Maximin criterion
- (C1) Formulating the decision problem for an optimal choice using expected value

Problem description

A coffee exporter must choose a shipping route. The transaction is exporting coffee shipments, where profit is earned per shipment delivered.

Decision alternatives: • **Route through Port A** • **Route through Port B**

Exchange-rate states: • Favorable rate (0.50) • Unfavorable rate (0.50)

Profit per shipment depends on the exchange rate: • Favorable → 6 thousand USD per shipment • Unfavorable → 2 thousand USD per shipment

Expected shipments: • Port A → 40 (Favorable), 28 (Unfavorable) • Port B → 32 (Favorable), 30 (Unfavorable)

Question: Compute profits and choose the best route using Maximin, Expected Value, and Maximax.

Solution

Structured solution (C2) Interpreting decision alternatives, events, consequences, and states +

Element	Description
Alternatives	Port A route; Port B route
States of nature	Favorable rate (0.50) Unfavorable rate (0.50)
Events	Exchange-rate level that affects profit per shipment
Consequences	Profit in thousand USD based on shipments × profit per shipment

(C3) Building the payoff table +

Profit calculations (thousand USD):

- Port A: Favorable $\rightarrow 40 \times 6 = 240$; Unfavorable $\rightarrow 28 \times 2 = 56$.
- Port B: Favorable $\rightarrow 32 \times 6 = 192$; Unfavorable $\rightarrow 30 \times 2 = 60$.

Route	Favorable rate (0.50)	Unfavorable rate (0.50)
Port A	240	56
Port B	192	60

(C4) Applying the Maximax criterion +

Maximax compares the best outcomes.

$$\begin{aligned}\max_i \text{Payoff}(\text{Port A}, S_i)\& &= \max\{240, 56\} = 240\\ \max_i \text{Payoff}(\text{Port B}, S_i)\& &= \max\{192, 60\} = 192\end{aligned}$$

$$\max\{240, 192\} = 240$$

The maximax choice is *routing through Port A*.

(C5) Applying the Maximin criterion +

Maximin compares the worst outcomes.

$$\begin{aligned}\min_i \text{Payoff}(\text{Port A}, S_i)\& &= \min\{240, 56\} = 56\\ \min_i \text{Payoff}(\text{Port B}, S_i)\& &= \min\{192, 60\} = 60\end{aligned}$$

$$\max\{56, 60\} = 60$$

The maximin recommendation is *routing through Port B*.

(C1) Formulating the decision problem for an optimal choice using expected value +

We choose the route that maximizes expected profit. Expected values:

$$\begin{aligned}\text{EV}_{\text{Port A}}\& &= 0.50(240) + 0.50(56) = 120 + 28 = 148\\ \text{EV}_{\text{Port B}}\& &= 0.50(192) + 0.50(60) = 96 + 30 = 126\end{aligned}$$

$$\max\{148, 126\} = 148$$

Summary of recommendations across decision criteria

- Maximax favors *Port A* (highest payoff of 240).
- Maximin favors *Port B* (best worst-case payoff of 60).
- Expected value favors *Port A* (EV of 148).

Port A offers higher upside and average profit, while Port B provides slightly better downside protection.

(C2) Interpreting decision alternatives, events, consequences, and states

Element	Description
Alternatives	Port A route; Port B route
States of nature	Favorable rate (0.50) Unfavorable rate (0.50)
Events	Exchange-rate level that affects profit per shipment
Consequences	Profit in thousand USD based on shipments $\times$ profit per shipment

(C3) Building the payoff table

Profit calculations (thousand USD):

- Port A: Favorable $\rightarrow 40 \times 6 = 240$; Unfavorable $\rightarrow 28 \times 2 = 56$.
- Port B: Favorable $\rightarrow 32 \times 6 = 192$; Unfavorable $\rightarrow 30 \times 2 = 60$.

Route	Favorable rate (0.50)	Unfavorable rate (0.50)
Port A	240	56
Port B	192	60

(C4) Applying the Maximax criterion

Maximax compares the best outcomes.

$$\begin{aligned}\max_i \text{Payoff}(\text{Port A}, S_i) &= \max\{240, 56\} = 240\\ \max_i \text{Payoff}(\text{Port B}, S_i) &= \max\{192, 60\} = 192\end{aligned}$$

$$\max\{240, 192\} = 240$$

The maximax choice is *routing through Port A*.

(C5) Applying the Maximin criterion

Maximin compares the worst outcomes.

$$\begin{aligned}\min_i \text{Payoff}(\text{Port A}, S_i) &= \min\{240, 56\} = 56\\ \min_i \text{Payoff}(\text{Port B}, S_i) &= \min\{192, 60\} = 60\end{aligned}$$

$$\max\{56, 60\} = 60$$

The maximin recommendation is *routing through Port B*.

(C1) Formulating the decision problem for an optimal choice using expected value

We choose the route that maximizes expected profit. Expected values:

$$\begin{aligned} \text{EV}_{\text{Port A}} &= 0.50(240) + 0.50(56) = 120 + 28 = 148 \\ \text{EV}_{\text{Port B}} &= 0.50(192) + 0.50(60) = 96 + 30 = 126 \end{aligned}$$

$$\max\{148, 126\} = 148$$

Summary of recommendations across decision criteria

- Maximax favors *Port A* (highest payoff of 240).
- Maximin favors *Port B* (best worst-case payoff of 60).
- Expected value favors *Port A* (EV of 148).

Port A offers higher upside and average profit, while Port B provides slightly better downside protection.

Problem 7 — Lake Ferry Tourism

- Problem description
- (C2) Interpreting decision alternatives, events, consequences, and states
- (C3) Building the payoff table
- (C4) Applying the Maximax criterion
- (C5) Applying the Maximin criterion
- (C1) Formulating the decision problem for an optimal choice using expected value

Problem description

A lake tourism operator is choosing boat sizes for sightseeing tours. The transaction is tour ticket sales, where profit is earned per ticket sold.

Alternatives: • **Operate small ferries** • **Operate large ferries**

Seasonal states: • Peak season (0.40) • Normal season (0.35) • Rainy season (0.25)

Profit per ticket depends on the season: • Peak → 0.05 thousand USD • Normal → 0.04 thousand USD • Rainy → 0.03 thousand USD

Expected tickets sold: • Small ferries → 2200 (Peak), 1600 (Normal), 1000 (Rainy) • Large ferries → 2600 (Peak), 1800 (Normal), 800 (Rainy)

Question: Compute profits and choose the best fleet size using Maximin, Expected Value, and Maximax.

Solution

Structured solution (C2) Interpreting decision alternatives, events, consequences, and states +

Element	Description
Alternatives	Small ferries; Large ferries
States of nature	Peak (0.40) Normal (0.35) Rainy (0.25)
Events	Tourism season and ticket price level
Consequences	Profit in thousand USD based on tickets × profit per ticket

(C3) Building the payoff table +

Profit calculations (thousand USD):

- Small ferries: Peak $\rightarrow 2200 \times 0.05 = 110$; Normal $\rightarrow 1600 \times 0.04 = 64$; Rainy $\rightarrow 1000 \times 0.03 = 30$.
- Large ferries: Peak $\rightarrow 2600 \times 0.05 = 130$; Normal $\rightarrow 1800 \times 0.04 = 72$; Rainy $\rightarrow 800 \times 0.03 = 24$.

Alternative	Peak (0.40)	Normal (0.35)	Rainy (0.25)
Small ferries	110	64	30
Large ferries	130	72	24

(C4) Applying the Maximax criterion +

Maximax compares the best outcomes.

$$\begin{aligned} \max_i \text{Payoff}(\text{Small}, S_i) &= \max\{110, 64, 30\} = 110 \\ \max_i \text{Payoff}(\text{Large}, S_i) &= \max\{130, 72, 24\} = 130 \end{aligned}$$

$$\max\{110, 130\} = 130$$

The maximax choice is *large ferries*.

(C5) Applying the Maximin criterion +

Maximin compares the worst outcomes.

$$\begin{aligned} \min_i \text{Payoff}(\text{Small}, S_i) &= \min\{110, 64, 30\} = 30 \\ \min_i \text{Payoff}(\text{Large}, S_i) &= \min\{130, 72, 24\} = 24 \end{aligned}$$

$$\max\{30, 24\} = 30$$

The maximin recommendation is *small ferries*.

(C1) Formulating the decision problem for an optimal choice using expected value +

We select the fleet size that maximizes expected profit. Expected values:

$$\begin{aligned} \mathrm{EV}_{\mathrm{Small}} &= 0.40(110) + 0.35(64) + 0.25(30) = 44 + 22.4 + 7.5 = 73.9 \\ \mathrm{EV}_{\mathrm{Large}} &= 0.40(130) + 0.35(72) + 0.25(24) = 52 + 25.2 + 6 = 83.2 \end{aligned}$$

$$\max\{73.9, 83.2\} = 83.2$$

Summary of recommendations across decision criteria

- Maximax favors *large ferries* (highest payoff of 130).
- Maximin favors *small ferries* (best worst-case payoff of 30).
- Expected value favors *large ferries* (EV of 83.2).

Large ferries provide higher upside and average profit, but small ferries protect the rainy-season downside.

(C2) Interpreting decision alternatives, events, consequences, and states

Element	Description
Alternatives	Small ferries; Large ferries
States of nature	Peak (0.40) Normal (0.35) Rainy (0.25)
Events	Tourism season and ticket price level
Consequences	Profit in thousand USD based on tickets $\times$ profit per ticket

(C3) Building the payoff table

Profit calculations (thousand USD):

- Small ferries: Peak $\rightarrow 2200 \times 0.05 = 110$; Normal $\rightarrow 1600 \times 0.04 = 64$; Rainy $\rightarrow 1000 \times 0.03 = 30$.
- Large ferries: Peak $\rightarrow 2600 \times 0.05 = 130$; Normal $\rightarrow 1800 \times 0.04 = 72$; Rainy $\rightarrow 800 \times 0.03 = 24$.

Alternative	Peak (0.40)	Normal (0.35)	Rainy (0.25)
Small ferries	110	64	30
Large ferries	130	72	24

(C4) Applying the Maximax criterion

Maximax compares the best outcomes.

$$\begin{aligned} \max_i \mathrm{Payoff}(\mathrm{Small}, S_i) &= \max\{110, 64, 30\} = 110 \\ \max_i \mathrm{Payoff}(\mathrm{Large}, S_i) &= \max\{130, 72, 24\} = 130 \end{aligned}$$

$$\max\{110, 130\} = 130$$

The maximax choice is *large ferries*.

(C5) Applying the Maximin criterion

Maximin compares the worst outcomes.

$$\begin{aligned}\min_i \text{Payoff}(\text{Small}, S_i)\& &= \min\{110, 64, 30\} = 30\\ \min_i \text{Payoff}(\text{Large}, S_i)\& &= \min\{130, 72, 24\} = 24\end{aligned}$$

$$\max\{30, 24\} = 30$$

The maximin recommendation is *small ferries*.

(C1) Formulating the decision problem for an optimal choice using expected value

We select the fleet size that maximizes expected profit. Expected values:

$$\begin{aligned}\text{EV}_{\text{Small}}\& &= 0.40(110) + 0.35(64) + 0.25(30) = 44 + 22.4 + 7.5 = 73.9\\ \text{EV}_{\text{Large}}\& &= 0.40(130) + 0.35(72) + 0.25(24) = 52 + 25.2 + 6 = 83.2\end{aligned}$$

$$\max\{73.9, 83.2\} = 83.2$$

Summary of recommendations across decision criteria

- Maximax favors *large ferries* (highest payoff of 130).
- Maximin favors *small ferries* (best worst-case payoff of 30).
- Expected value favors *large ferries* (EV of 83.2).

Large ferries provide higher upside and average profit, but small ferries protect the rainy-season downside.

Problem 8 — Cold-Storage Contracts

- Problem description
- (C2) Interpreting decision alternatives, events, consequences, and states
- (C3) Building the payoff table
- (C4) Applying the Maximax criterion
- (C5) Applying the Maximin criterion
- (C1) Formulating the decision problem for an optimal choice using expected value

Problem description

A logistics company is leasing cold-storage space for agricultural pallets. The transaction is storage contracts, where profit is earned per pallet stored plus a fixed profit associated with the chosen facility.

Decision alternatives: • **Small facility** (fixed profit 40 thousand USD, profit 0.18 thousand USD per pallet) • **Medium facility** (fixed profit 25 thousand USD, profit 0.12 thousand USD per pallet) • **Large facility** (fixed profit 10 thousand USD, profit 0.08 thousand USD per pallet)

Demand states: • High storage demand (0.55) • Low storage demand (0.45)

Expected pallets stored: • Small → 300 (High), 220 (Low) • Medium → 600 (High), 420 (Low) • Large → 900 (High), 700 (Low)

Question: Compute profits and choose the best facility using Maximin, Expected Value, and Maximax.

Solution

Structured solution (C2) Interpreting decision alternatives, events, consequences, and states +

Element	Description
Alternatives	Small facility; Medium facility; Large facility
States of nature	High demand (0.55) Low demand (0.45)
Events	Storage demand realized after the lease decision
Consequences	Profit in thousand USD = pallets × profit per pallet + fixed profit

(C3) Building the payoff table +

Profit calculations (thousand USD):

- Small: High → $300 \times 0.18 + 40 = 94$; Low → $220 \times 0.18 + 40 = 79.6$.
- Medium: High → $600 \times 0.12 + 25 = 97$; Low → $420 \times 0.12 + 25 = 75.4$.
- Large: High → $900 \times 0.08 + 10 = 82$; Low → $700 \times 0.08 + 10 = 66$.

Alternative	High demand (0.55)	Low demand (0.45)
Small facility	94	79.6
Medium facility	97	75.4
Large facility	82	66

(C4) Applying the Maximax criterion +

Maximax compares the best outcomes.

$$\begin{aligned} \max_i \text{Payoff}(\text{Small}, S_i) &= \max\{94, 79.6\} = 94 \\ \max_i \text{Payoff}(\text{Medium}, S_i) &= \max\{97, 75.4\} = 97 \\ \max_i \text{Payoff}(\text{Large}, S_i) &= \max\{82, 66\} = 82 \end{aligned}$$

$$\max\{94, 97, 82\} = 97$$

The maximax choice is the *medium facility*.

(C5) Applying the Maximin criterion +

Maximin compares the worst outcomes.

$$\begin{aligned}
\min_i \text{Payoff}(\text{Small}, S_i) &= \min\{94, 79.6\} = 79.6 \\
\min_i \text{Payoff}(\text{Medium}, S_i) &= \min\{97, 75.4\} = 75.4 \\
\min_i \text{Payoff}(\text{Large}, S_i) &= \min\{82, 66\} = 66
\end{aligned}$$

$$\max\{79.6, 75.4, 66\} = 79.6$$

The maximin recommendation is the *small facility*.

(C1) Formulating the decision problem for an optimal choice using expected value +

We select the facility that maximizes expected profit. Expected values:

$$\begin{aligned}
\text{EV}_{\text{Small}} &= 0.55(94) + 0.45(79.6) = 51.7 + 35.82 = 87.52 \\
\text{EV}_{\text{Medium}} &= 0.55(97) + 0.45(75.4) = 53.35 + 33.93 = 87.28 \\
\text{EV}_{\text{Large}} &= 0.55(82) + 0.45(66) = 45.1 + 29.7 = 74.8
\end{aligned}$$

$$\max\{87.52, 87.28, 74.8\} = 87.52$$

Summary of recommendations across decision criteria

- Maximax favors the *medium facility* (highest payoff of 97).
- Maximin favors the *small facility* (best worst-case payoff of 79.6).
- Expected value favors the *small facility* (EV of 87.52).

The medium facility has the best upside, while the small facility provides the best downside protection and slightly higher expected profit.

(C2) Interpreting decision alternatives, events, consequences, and states

Element	Description
Alternatives	Small facility; Medium facility; Large facility
States of nature	High demand (0.55) Low demand (0.45)
Events	Storage demand realized after the lease decision
Consequences	Profit in thousand USD = pallets × profit per pallet + fixed profit

(C3) Building the payoff table

Profit calculations (thousand USD):

- Small: High $\rightarrow 300 \times 0.18 + 40 = 94$; Low $\rightarrow 220 \times 0.18 + 40 = 79.6$.
- Medium: High $\rightarrow 600 \times 0.12 + 25 = 97$; Low $\rightarrow 420 \times 0.12 + 25 = 75.4$.
- Large: High $\rightarrow 900 \times 0.08 + 10 = 82$; Low $\rightarrow 700 \times 0.08 + 10 = 66$.

Alternative	High demand (0.55)	Low demand (0.45)
Small facility	94	79.6
Medium facility	97	75.4
Large facility	82	66

(C4) Applying the Maximax criterion

Maximax compares the best outcomes.

$$\begin{aligned} \max_i \text{Payoff}(\text{Small}, S_i) &= \max\{94, 79.6\} = 94 \\ \max_i \text{Payoff}(\text{Medium}, S_i) &= \max\{97, 75.4\} = 97 \\ \max_i \text{Payoff}(\text{Large}, S_i) &= \max\{82, 66\} = 82 \end{aligned}$$

$$\max\{94, 97, 82\} = 97$$

The maximax choice is the *medium facility.*

(C5) Applying the Maximin criterion

Maximin compares the worst outcomes.

$$\begin{aligned} \min_i \text{Payoff}(\text{Small}, S_i) &= \min\{94, 79.6\} = 79.6 \\ \min_i \text{Payoff}(\text{Medium}, S_i) &= \min\{97, 75.4\} = 75.4 \\ \min_i \text{Payoff}(\text{Large}, S_i) &= \min\{82, 66\} = 66 \end{aligned}$$

$$\max\{79.6, 75.4, 66\} = 79.6$$

The maximin recommendation is the *small facility.*

(C1) Formulating the decision problem for an optimal choice using expected value

We select the facility that maximizes expected profit. Expected values:

$$\begin{aligned} \text{EV}_{\text{Small}} &= 0.55(94) + 0.45(79.6) = 51.7 + 35.82 = 87.52 \\ \text{EV}_{\text{Medium}} &= 0.55(97) + 0.45(75.4) = 53.35 + 33.93 = 87.28 \\ \text{EV}_{\text{Large}} &= 0.55(82) + 0.45(66) = 45.1 + 29.7 = 74.8 \end{aligned}$$

$$\max\{87.52, 87.28, 74.8\} = 87.52$$

Summary of recommendations across decision criteria

- Maximax favors the *medium facility* (highest payoff of 97).
- Maximin favors the *small facility* (best worst-case payoff of 79.6).
- Expected value favors the *small facility* (EV of 87.52).

The medium facility has the best upside, while the small facility provides the best downside protection and slightly higher expected profit.

Problem 9 — Solar Installation Bids

- Problem description
- (C2) Interpreting decision alternatives, events, consequences, and states
- (C3) Building the payoff table
- (C4) Applying the Maximax criterion
- (C5) Applying the Maximin criterion
- (C1) Formulating the decision problem for an optimal choice using expected value

Problem description

A solar company is bidding on installation contracts. The transaction is installing solar systems, where profit is earned per installation plus a fixed profit adjustment from policy conditions.

Decision alternatives: • **Residential focus** (profit 0.9 thousand USD per installation) • **Commercial focus** (profit 1.6 thousand USD per installation) • **Mixed portfolio** (profit 1.2 thousand USD per installation)

Policy states with fixed profit components: • Rebate continues (0.55) $\rightarrow$ +20 thousand USD • Rebate ends (0.45) $\rightarrow$ -10 thousand USD

Expected installations: • Residential $\rightarrow$ 80 (Rebate), 50 (No rebate) • Commercial $\rightarrow$ 55 (Rebate), 40 (No rebate) • Mixed $\rightarrow$ 70 (Rebate), 45 (No rebate)

Question: Compute profits and choose the best bid strategy using Maximin, Expected Value, and Maximax.

Solution

Structured solution (C2) Interpreting decision alternatives, events, consequences, and states +

Element	Description
Alternatives	Residential; Commercial; Mixed portfolio
States of nature	Rebate continues (0.55) Rebate ends (0.45)
Events	Policy outcome affecting the fixed profit adjustment
Consequences	Profit in thousand USD = installs $\times$ profit per install + fixed profit

(C3) Building the payoff table +

Profit calculations (thousand USD):

- Residential: Rebate $\rightarrow 80 \times 0.9 + 20 = 92$; No rebate $\rightarrow 50 \times 0.9 - 10 = 35$.
- Commercial: Rebate $\rightarrow 55 \times 1.6 + 20 = 108$; No rebate $\rightarrow 40 \times 1.6 - 10 = 54$.
- Mixed: Rebate $\rightarrow 70 \times 1.2 + 20 = 104$; No rebate $\rightarrow 45 \times 1.2 - 10 = 44$.

Alternative	Rebate continues (0.55)	Rebate ends (0.45)
Residential focus	92	35
Commercial focus	108	54
Mixed portfolio	104	44

(C4) Applying the Maximax criterion +

Maximax compares the best outcomes.

$$\begin{aligned} \max_i \text{Payoff}(\text{Residential}, S_i) &= \max\{92, 35\} = 92 \\ \max_i \text{Payoff}(\text{Commercial}, S_i) &= \max\{108, 54\} = 108 \\ \max_i \text{Payoff}(\text{Mixed}, S_i) &= \max\{104, 44\} = 104 \end{aligned}$$

$$\max\{92, 108, 104\} = 108$$

The maximax choice is the *commercial focus*.

(C5) Applying the Maximin criterion +

Maximin compares the worst outcomes.

$$\begin{aligned} \min_i \text{Payoff}(\text{Residential}, S_i) &= \min\{92, 35\} = 35 \\ \min_i \text{Payoff}(\text{Commercial}, S_i) &= \min\{108, 54\} = 54 \\ \min_i \text{Payoff}(\text{Mixed}, S_i) &= \min\{104, 44\} = 44 \end{aligned}$$

$$\max\{35, 54, 44\} = 54$$

The maximin recommendation is the *commercial focus*.

(C1) Formulating the decision problem for an optimal choice using expected value +

We select the bid strategy that maximizes expected profit. Expected values:

$$\begin{aligned} \text{EV}_{\text{Residential}} &= 0.55(92) + 0.45(35) = 50.6 + 15.75 = 66.35 \\ \text{EV}_{\text{Commercial}} &= 0.55(108) + 0.45(54) = 59.4 + 24.3 = 83.7 \\ \text{EV}_{\text{Mixed}} &= 0.55(104) + 0.45(44) = 57.2 + 19.8 = 77 \end{aligned}$$

$$\max\{66.35, 83.7, 77\} = 83.7$$

Summary of recommendations across decision criteria

- Maximax favors the *commercial focus* (highest payoff of 108).
- Maximin favors the *commercial focus* (best worst-case payoff of 54).
- Expected value favors the *commercial focus* (EV of 83.7).

The commercial focus leads on upside, downside protection, and average profit when the rebate is uncertain.

(C2) Interpreting decision alternatives, events, consequences, and states

Element	Description
Alternatives	Residential; Commercial; Mixed portfolio
States of nature	Rebate continues (0.55) Rebate ends (0.45)
Events	Policy outcome affecting the fixed profit adjustment
Consequences	Profit in thousand USD = installs $\times$ profit per install + fixed profit

(C3) Building the payoff table

Profit calculations (thousand USD):

- Residential: Rebate $\rightarrow 80 \times 0.9 + 20 = 92$; No rebate $\rightarrow 50 \times 0.9 - 10 = 35$.
- Commercial: Rebate $\rightarrow 55 \times 1.6 + 20 = 108$; No rebate $\rightarrow 40 \times 1.6 - 10 = 54$.
- Mixed: Rebate $\rightarrow 70 \times 1.2 + 20 = 104$; No rebate $\rightarrow 45 \times 1.2 - 10 = 44$.

Alternative	Rebate continues (0.55)	Rebate ends (0.45)
Residential focus	92	35
Commercial focus	108	54
Mixed portfolio	104	44

(C4) Applying the Maximax criterion

Maximax compares the best outcomes.

$$\begin{aligned}
\max_i \text{Payoff}(\text{Residential}, S_i) &= \max\{92, 35\} = 92 \\
\max_i \text{Payoff}(\text{Commercial}, S_i) &= \max\{108, 54\} = 108 \\
\max_i \text{Payoff}(\text{Mixed}, S_i) &= \max\{104, 44\} = 104
\end{aligned}$$

$$\max\{92, 108, 104\} = 108$$

The maximax choice is the *commercial focus*.

(C5) Applying the Maximin criterion

Maximin compares the worst outcomes.

$$\begin{aligned}\min_i \text{Payoff}(\text{Residential}, S_i) &= \min\{92, 35\} = 35\\ \min_i \text{Payoff}(\text{Commercial}, S_i) &= \min\{108, 54\} = 54\\ \min_i \text{Payoff}(\text{Mixed}, S_i) &= \min\{104, 44\} = 44\end{aligned}$$

$$\max\{35, 54, 44\} = 54$$

The maximin recommendation is the *commercial focus*.

(C1) Formulating the decision problem for an optimal choice using expected value

We select the bid strategy that maximizes expected profit. Expected values:

$$\begin{aligned}\text{EV}_{\text{Residential}} &= 0.55(92) + 0.45(35) = 50.6 + 15.75 = 66.35\\ \text{EV}_{\text{Commercial}} &= 0.55(108) + 0.45(54) = 59.4 + 24.3 = 83.7\\ \text{EV}_{\text{Mixed}} &= 0.55(104) + 0.45(44) = 57.2 + 19.8 = 77\end{aligned}$$

$$\max\{66.35, 83.7, 77\} = 83.7$$

Summary of recommendations across decision criteria

- Maximax favors the *commercial focus* (highest payoff of 108).
- Maximin favors the *commercial focus* (best worst-case payoff of 54).
- Expected value favors the *commercial focus* (EV of 83.7).

The commercial focus leads on upside, downside protection, and average profit when the rebate is uncertain.

Problem 10 — Air Cargo Charters

- Problem description
- (C2) Interpreting decision alternatives, events, consequences, and states
- (C3) Building the payoff table
- (C4) Applying the Maximax criterion
- (C5) Applying the Maximin criterion
- (C1) Formulating the decision problem for an optimal choice using expected value

Problem description

An air cargo firm is choosing a charter fleet. The transaction is cargo charters, where profit is earned per charter plus a fixed profit component tied to the chosen aircraft lease.

Alternatives: • **Lease narrow-body aircraft** (fixed profit 30 thousand USD) • **Lease wide-body aircraft** (fixed profit 50 thousand USD)

Fuel-price states: • Low fuel cost (0.30) • Medium fuel cost (0.45) • High fuel cost (0.25)

Profit per charter depends on fuel prices: • Low fuel → 8 thousand USD • Medium fuel → 5 thousand USD • High fuel → 2 thousand USD

Expected charters: • Narrow-body → 20 (Low), 18 (Medium), 15 (High) • Wide-body → 14 (Low), 12 (Medium), 9 (High)

Question: Compute profits and choose the best lease using Maximin, Expected Value, and Maximax.

Solution

Structured solution (C2) Interpreting decision alternatives, events, consequences, and states +

Element	Description
Alternatives	Narrow-body lease; Wide-body lease
States of nature	Low fuel (0.30) Medium fuel (0.45) High fuel (0.25)
Events	Fuel prices affecting profit per charter
Consequences	Profit in thousand USD = charters × profit per charter + fixed profit

(C3) Building the payoff table +

Profit calculations (thousand USD):

- Narrow-body: Low → $20\times8+30=190$; Medium → $18\times5+30=120$; High → $15\times2+30=60$.
- Wide-body: Low → $14\times8+50=162$; Medium → $12\times5+50=110$; High → $9\times2+50=68$.

Alternative	Low fuel (0.30)	Medium fuel (0.45)	High fuel (0.25)
Narrow-body lease	190	120	60
Wide-body lease	162	110	68

(C4) Applying the Maximax criterion +

Maximax compares the best outcomes.

$$\begin{aligned}\max_i \text{Payoff}(\text{Narrow}, S_i) &= \max\{190, 120, 60\} = 190\\ \max_i \text{Payoff}(\text{Wide}, S_i) &= \max\{162, 110, 68\} = 162\end{aligned}$$

$$\max\{190, 162\} = 190$$

The maximax choice is the *narrow-body lease*.

(C5) Applying the Maximin criterion +

Maximin compares the worst outcomes.

$$\min_i \text{Payoff}(\text{Narrow}, S_i)\& = \min\{190, 120, 60\} = 60$$
$$\min_i \text{Payoff}(\text{Wide}, S_i)\& = \min\{162, 110, 68\} = 68$$
$$\max\{60, 68\} = 68$$

The maximin recommendation is the *wide-body lease*.

(C1) Formulating the decision problem for an optimal choice using expected value +

We select the lease that maximizes expected profit. Expected values:

$$\text{EV}_{\text{Narrow}}\& = 0.30(190) + 0.45(120) + 0.25(60) = 57 + 54 + 15 = 126$$
$$\text{EV}_{\text{Wide}}\& = 0.30(162) + 0.45(110) + 0.25(68) = 48.6 + 49.5 + 17 = 115.1$$
$$\max\{126, 115.1\} = 126$$

Summary of recommendations across decision criteria

- Maximax favors the *narrow-body lease* (highest payoff of 190).
- Maximin favors the *wide-body lease* (best worst-case payoff of 68).
- Expected value favors the *narrow-body lease* (EV of 126).

Narrow-body aircraft provide stronger upside and expected profit, while wide-body leases offer better downside protection when fuel costs spike.

(C2) Interpreting decision alternatives, events, consequences, and states

Element	Description
Alternatives	Narrow-body lease; Wide-body lease
States of nature	Low fuel (0.30) Medium fuel (0.45) High fuel (0.25)
Events	Fuel prices affecting profit per charter
Consequences	Profit in thousand USD = charters × profit per charter + fixed profit

(C3) Building the payoff table

Profit calculations (thousand USD):

- Narrow-body: Low → $20\times8+30 = 190$; Medium → $18\times5+30 = 120$; High → $15\times2+30 = 60$.
- Wide-body: Low → $14\times8+50 = 162$; Medium → $12\times5+50 = 110$; High → $9\times2+50 = 68$.

Alternative	Low fuel (0.30)	Medium fuel (0.45)	High fuel (0.25)
Narrow-body lease	190	120	60
Wide-body lease	162	110	68

(C4) Applying the Maximax criterion

Maximax compares the best outcomes.

$$\begin{aligned}\max_i \text{Payoff}(\text{Narrow}, S_i) &= \max\{190, 120, 60\} = 190\\ \max_i \text{Payoff}(\text{Wide}, S_i) &= \max\{162, 110, 68\} = 162\end{aligned}$$

$$\max\{190, 162\} = 190$$

The maximax choice is the *narrow-body lease*.

(C5) Applying the Maximin criterion

Maximin compares the worst outcomes.

$$\begin{aligned}\min_i \text{Payoff}(\text{Narrow}, S_i) &= \min\{190, 120, 60\} = 60\\ \min_i \text{Payoff}(\text{Wide}, S_i) &= \min\{162, 110, 68\} = 68\end{aligned}$$

$$\max\{60, 68\} = 68$$

The maximin recommendation is the *wide-body lease*.

(C1) Formulating the decision problem for an optimal choice using expected value

We select the lease that maximizes expected profit. Expected values:

$$\begin{aligned}\text{EV}_{\text{Narrow}} &= 0.30(190) + 0.45(120) + 0.25(60) = 57 + 54 + 15 = 126\\ \text{EV}_{\text{Wide}} &= 0.30(162) + 0.45(110) + 0.25(68) = 48.6 + 49.5 + 17 = 115.1\end{aligned}$$

$$\max\{126, 115.1\} = 126$$

Summary of recommendations across decision criteria

- Maximax favors the *narrow-body lease* (highest payoff of 190).
- Maximin favors the *wide-body lease* (best worst-case payoff of 68).
- Expected value favors the *narrow-body lease* (EV of 126).

Narrow-body aircraft provide stronger upside and expected profit, while wide-body leases offer better downside protection when fuel costs spike.

Problem 11 — Seafood Export Licenses

- Problem description
- (C2) Interpreting decision alternatives, events, consequences, and states
- (C3) Building the payoff table
- (C4) Applying the Maximax criterion
- (C5) Applying the Maximin criterion
- (C1) Formulating the decision problem for an optimal choice using expected value

Problem description

A seafood exporter is choosing a license focus. The transaction is exporting seafood crates, where profit is earned per crate plus a fixed profit adjustment tied to seasonal conditions.

Alternatives: • **Frozen fillet focus** • **Fresh export focus**

Seasonal states with fixed profit components: • Strong tourism season (0.40) → +30 thousand USD • Normal season (0.35) → +10 thousand USD • Stormy season (0.25) → −15 thousand USD

Profit per crate depends on the season: • Strong → 0.7 thousand USD • Normal → 0.5 thousand USD • Stormy → 0.2 thousand USD

Expected crates exported: • Frozen fillet → 120 (Strong), 100 (Normal), 80 (Stormy) • Fresh export → 90 (Strong), 110 (Normal), 70 (Stormy)

Question: Compute profits and choose the best focus using Maximin, Expected Value, and Maximax.

Solution

Structured solution (C2) Interpreting decision alternatives, events, consequences, and states +

Element	Description
Alternatives	Frozen fillet focus; Fresh export focus
States of nature	Strong (0.40) Normal (0.35) Stormy (0.25)
Events	Seasonal conditions affecting prices and fixed profit adjustments
Consequences	Profit in thousand USD = crates × profit per crate + fixed profit

(C3) Building the payoff table +

Profit calculations (thousand USD):

- Frozen fillet: Strong → $120 \times 0.7 + 30 = 114$; Normal → $100 \times 0.5 + 10 = 60$; Stormy → $80 \times 0.2 - 15 = 1$.

- Fresh export: Strong $\rightarrow 90 \times 0.7 + 30 = 93$; Normal $\rightarrow 110 \times 0.5 + 10 = 65$; Stormy $\rightarrow 70 \times 0.2 - 15 = -1$.

Alternative	Strong (0.40)	Normal (0.35)	Stormy (0.25)
Frozen fillet focus	114	60	1
Fresh export focus	93	65	-1

(C4) Applying the Maximax criterion +

Maximax compares the best outcomes.

$$\begin{aligned} \max_i \text{Payoff}(\text{Frozen}, S_i) &= \max\{114, 60, 1\} = 114 \\ \max_i \text{Payoff}(\text{Fresh}, S_i) &= \max\{93, 65, -1\} = 93 \end{aligned}$$

$$\max\{114, 93\} = 114$$

The maximax choice is the *frozen fillet focus*.

(C5) Applying the Maximin criterion +

Maximin compares the worst outcomes.

$$\begin{aligned} \min_i \text{Payoff}(\text{Frozen}, S_i) &= \min\{114, 60, 1\} = 1 \\ \min_i \text{Payoff}(\text{Fresh}, S_i) &= \min\{93, 65, -1\} = -1 \end{aligned}$$

$$\max\{1, -1\} = 1$$

The maximin recommendation is the *frozen fillet focus*.

(C1) Formulating the decision problem for an optimal choice using expected value +

We select the license focus that maximizes expected profit. Expected values:

$$\begin{aligned} \text{EV}_{\text{Frozen}} &= 0.40(114) + 0.35(60) + 0.25(1) = 45.6 + 21 + 0.25 = 66.85 \\ \text{EV}_{\text{Fresh}} &= 0.40(93) + 0.35(65) + 0.25(-1) = 37.2 + 22.75 - 0.25 = 59.7 \end{aligned}$$

$$\max\{66.85, 59.7\} = 66.85$$

Summary of recommendations across decision criteria

- Maximax favors the *frozen fillet focus* (highest payoff of 114).
- Maximin favors the *frozen fillet focus* (best worst-case payoff of 1).
- Expected value favors the *frozen fillet focus* (EV of 66.85).

Frozen exports provide stronger upside and downside protection, leading to the highest average profit.

(C2) Interpreting decision alternatives, events, consequences, and states

Element	Description
Alternatives	Frozen fillet focus; Fresh export focus
States of nature	Strong (0.40) Normal (0.35) Stormy (0.25)
Events	Seasonal conditions affecting prices and fixed profit adjustments
Consequences	Profit in thousand USD = crates × profit per crate + fixed profit

(C3) Building the payoff table

Profit calculations (thousand USD):

- Frozen fillet: Strong $\rightarrow$ $120 \times 0.7 + 30 = 114$; Normal $\rightarrow$ $100 \times 0.5 + 10 = 60$; Stormy $\rightarrow$ $80 \times 0.2 - 15 = 1$.
- Fresh export: Strong $\rightarrow$ $90 \times 0.7 + 30 = 93$; Normal $\rightarrow$ $110 \times 0.5 + 10 = 65$; Stormy $\rightarrow$ $70 \times 0.2 - 15 = -1$.

Alternative	Strong (0.40)	Normal (0.35)	Stormy (0.25)
Frozen fillet focus	114	60	1
Fresh export focus	93	65	-1

(C4) Applying the Maximax criterion

Maximax compares the best outcomes.

$$\begin{aligned} \max_i \text{Payoff}(\text{Frozen}, S_i) &= \max\{114, 60, 1\} = 114 \\ \max_i \text{Payoff}(\text{Fresh}, S_i) &= \max\{93, 65, -1\} = 93 \end{aligned}$$

$$\max\{114, 93\} = 114$$

The maximax choice is the *frozen fillet focus*.

(C5) Applying the Maximin criterion

Maximin compares the worst outcomes.

$$\begin{aligned} \min_i \text{Payoff}(\text{Frozen}, S_i) &= \min\{114, 60, 1\} = 1 \\ \min_i \text{Payoff}(\text{Fresh}, S_i) &= \min\{93, 65, -1\} = -1 \end{aligned}$$

$$\max\{1, -1\} = 1$$

The maximin recommendation is the *frozen fillet focus*.

(C1) Formulating the decision problem for an optimal choice using expected value

We select the license focus that maximizes expected profit. Expected values:

$$\begin{aligned} \text{EV}_{\text{Frozen}} &= 0.40(114) + 0.35(60) + 0.25(1) = 45.6 + 21 + 0.25 = 66.85 \\ \text{EV}_{\text{Fresh}} &= 0.40(93) + 0.35(65) + 0.25(-1) = 37.2 + 22.75 - 0.25 = 59.7 \end{aligned}$$

$$\max\{66.85, 59.7\} = 66.85$$

Summary of recommendations across decision criteria

- Maximax favors the *frozen fillet focus* (highest payoff of 114).
- Maximin favors the *frozen fillet focus* (best worst-case payoff of 1).
- Expected value favors the *frozen fillet focus* (EV of 66.85).

Frozen exports provide stronger upside and downside protection, leading to the highest average profit.